

Rock Evidence Lab: Look Before You Label

BUILD • Science • 40 minutes

I can compare rock specimens using observed properties and create a group using a stated rule.

Prepare before pupils arrive

Setup: Code and pre-inspect each educational specimen. Keep names hidden at first. Prepare macro image parity, white trays, hand lenses, feature tags and two group mats.

Safety: Exclude unknown field, cliff, quarry or demolition samples. Remove loose, dusty or sharp specimens. Keep a complete no-touch image route and follow local hygiene controls.

Equipment: Six teacher-pre-screened coded educational rock specimens or macro cards; white trays; hand lenses; feature tags; two hoops or sorting mats; Rock Comparison Ticket.

Safety: Use only pre-inspected educational samples. Keep them on trays and below shoulder height. No throwing, tasting, smelling, grinding, scratching or deliberate breaking. Wash hands after handling.

Fallback: Use macro image cards, point while an adult moves a specimen, or direct the whole inspection. Touching rock is not required.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Compare and group rocks by their appearance and simple properties.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 4 min	Name the label basis kept the same in W8A. Define property as something we can observe or test.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 5 min	property — a feature we can notice or test observe — look closely and record classify — put into groups with a rule Keep the code and hide the name. Place one specimen on a white tray and observe it from more than one angle.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Inspect a second coded specimen using the same feature words. State one similarity and one difference.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Place each coded observation under its strongest named visible feature. The name is not the target.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Inspect or direct the inspection of coded rock specimens, compare two using the same features and create two groups with a stated rule. Record: A Rock Comparison Ticket with two codes, two observed properties, one similarity, one difference and one explicit grouping rule.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Compare two coded specimens and defend one group using observed properties. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move; written closure is optional when Voice and Audience are observable.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Several large crystals are visible in the speckled surface.

Yes. Another person can inspect the specimen and check whether several visible crystals are present.

Misconception repair

Repaired. Colour is one observation; several properties give a stronger comparison, but appearance still may not prove identity.

Guided checkpoint

Yes. Another person can inspect the specimen and check whether several visible crystals are present. Choose the line that says what can be seen, not a guessed name or use.

Code A

VISIBLE CRYSTALS. Code A has several visible crystals as its strongest supplied feature. Look for the interlocking shapes named in the observation.

Code B

FINE • NO LARGE FEATURE. Code B is recorded as fine-looking under this appearance rule. Do not invent a large feature when the observation says none is visible.

Code C

VISIBLE GRAINS. Code C is placed by its many visible grains; its faint bands can support a later regrouping. Choose the feature described as many and clearly visible.

Code D

FINE • NO LARGE FEATURE. Code D is recorded as mostly fine-looking in the supplied view. Use the visible evidence; do not use a hidden rock name.

Code E

BANDS / FLAT PLANES. Code E is placed with visible flat sheet-like planes. The strongest stated feature is a set of clear flat planes.

Code F

LARGE HOLES. Code F is placed by its many large visible holes. Match the feature that dominates the supplied view.

Mystery image T

BANDS / FLAT PLANES. Mystery T provides visible-band evidence for the bands/planes group. Use the bands stated in the image description, not a name guess.

Model limitation

Appearance alone cannot prove an unknown rock's identity or hidden properties. Lighting, scale, weathering and natural variation can change how a specimen looks.

Independent evidence

A Rock Comparison Ticket with two codes, two observed properties, one similarity, one difference and one explicit grouping rule.

Supported: Compare two coded macro cards with a four-word feature bank, then place them using one shown rule. Use: I can see ___. Both have ___ / only ___ has ___. Adult exact scribing is valid.

Standard: Observe four to six coded specimens, compare a pair and create two groups using one stated property rule. Record the rule before moving specimens. Use only evidence that another person can check.

Stretch: Create two different valid groupings, explain why both work and state why neither proves every specimen's identity. Separate classification under a rule from identification of an unknown rock.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_B_W9A_Rock_Evidence_Explore-1.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.